

NAME: _____

PERIOD: _____

What Can These Data Support?

AP Statistics · Unit 1 · Topic 1.1

[STAR] THE PROBLEM

Suppose a school newspaper records last night's sleep for 12 students in one class. All 12 values are valid. For this class, the school flags a sleep concern if more than 20% slept fewer than 6 hours.

Claim 1: “The full table shows a sleep concern in this class.”

Claim 2: “Leave out the two lowest sleep values and there is no sleep concern.”

Sleep last night

Student	Sleep (hours)
Aiden	5.0
Bea	7.0
Cruz	8.5
Dev	6.5
Elise	8.0
Fiona	7.5
Gabe	5.5
Hana	9.0
Imani	7.0
Jalen	6.0
Kira	8.0
Leo	4.5

FIRST TAKE — one sentence before you work the parts

Which claim, if either, would you print? Name one detail behind your choice.

[BOOK] USE DATA IN CONTEXT (keep on the page)

A variable is a characteristic recorded for each individual. Variation means the values differ. A threshold is a cutoff used to decide whether action is needed. Percent = $100 \times \frac{\text{count meeting the condition}}{\text{total count}}$. When observations are removed, count the remaining observations and use the remaining total. Interpret the result in the original context.

Work (a)–(c).

DOK 1 (a) Name the variable and its unit.

DOK 2 (b) Find the percent who slept fewer than 6 hours using all 12 students. Then leave out the two lowest values and find the new percent.

DOK 3 (c) Which claim would you print? Use both percents and the school's rule. Explain why keeping or removing the two values changes what readers learn about this class.

Frame: I would print Claim _____ because _____% is _____ the school's cutoff.

Frame: Removing the two values changes _____, so readers would think _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored **E / P / I**.